

МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ

НАЦІОНАЛЬНИЙ ТЕХНІЧНИЙ УНІВЕРСИТЕТ
«ХАРКІВСЬКИЙ ПОЛІТЕХНІЧНИЙ ІНСТИТУТ»

Звіт

з лабораторної роботи № 5

з дисципліни «Стек технологій проєктування WEB-застосунків»

Виконав:

студент групи ІКМ-М225В

Суліма Нікіта Володимирович

Перевірив:

аспірант каф. КМПС Хорошун Андрій Сергійович

Харків 2026

Containers

CommentController

```
<?php

namespace App\Http\Controllers;

use App\Models\Project;
use Illuminate\Http\Request;

class ProjectController extends Controller
{
    // GET /api/projects
    public function index()
    {
        return response()->json(
            Project::with('tasks')->get()
        );
    }

    // POST /api/projects
    public function store(Request $request)
    {
        $data = $request->validate([
            'name' => 'required',
            'description' => 'nullable',
            'user_id' => 'required|exists:users,id'
        ]);

        $project = Project::create($data);

        return response()->json($project, 201);
    }
}
```

Controller

```
<?php

namespace App\Http\Controllers;

abstract class Controller
{
    //
}
```

ProjectController

```
<?php

namespace App\Http\Controllers;

use App\Models\Project;
use Illuminate\Http\Request;

class ProjectController extends Controller
{
    public function index()
    {
        return response()->json(
            Project::with('tasks')->get()
        );
    }

    public function store(Request $request)
    {
        $data = $request->validate([
            'name' => 'required',
            'description' => 'nullable',
            'user_id' => 'required|exists:users,id'
        ]);
    }
}
```

```
$project = Project::create($data);

return response()->json($project, 201);
}
}
```

TaskController

```
<?php

namespace App\Http\Controllers;

use App\Models\Task;
use Illuminate\Http\Request;

class TaskController extends Controller
{
    public function index()
    {
        return response()-
>json(Task::with(['project','comments','author','assignedTo'])->get());
    }

    public function store(Request $request)
    {
        $data = $request->validate([
            'title' => 'required',
            'description' => 'nullable',
            'status' => 'nullable',
            'project_id' => 'required|exists:projects,id',
            'author_id' => 'required|exists:users,id',
            'assigned_to' => 'nullable|exists:users,id',
            'due_date' => 'nullable|date'
        ]);
    }
}
```

```

        $task = Task::create($data);

        return response()->json($task, 201);
    }

    public function show($id)
    {
        return response()->json(Task::with(['project','comments'])-
>findOrFail($id));
    }

    public function update(Request $request, $id)
    {
        $task = Task::findOrFail($id);

        $task->update($request->all());

        return response()->json($task);
    }

    public function destroy($id)
    {
        Task::destroy($id);

        return response()->json(['message' => 'Deleted']);
    }
}

```

Model

Comment

```
<?php
```

```
namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use App\Models\Task;
use App\Models\User;

class Comment extends Model
{
    protected $fillable = [
        'task_id',
        'user_id',
        'content'
    ];

    public function task()
    {
        return $this->belongsTo(Task::class);
    }

    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

File

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class File extends Model
{
```

```
protected $fillable = [  
    'file_path',  
    'original_name',  
    'fileable_id',  
    'fileable_type'  
];  
  
public function fileable()  
{  
    return $this->morphTo();  
}  
}
```

Projects

```
<?php  
  
namespace App\Models;  
  
use Illuminate\Database\Eloquent\Model;  
use App\Models\User;  
use App\Models\Task;  
  
class Project extends Model  
{  
    protected $fillable = [  
        'name',  
        'description',  
        'user_id'  
    ];  
  
    public function user()  
    {  
        return $this->belongsTo(User::class);  
    }  
}
```

```
public function tasks()
{
    return $this->hasMany(Task::class);
}
}
```

Task

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use App\Models\Project;
use App\Models\Comment;
use App\Models\User;

class Task extends Model
{
    protected $fillable = [
        'title',
        'description',
        'status',
        'project_id',
        'author_id',
        'assigned_to',
        'due_date'
    ];

    // Task → Project
    public function project()
    {
        return $this->belongsTo(Project::class);
    }
}
```

```

// Task → Comments
public function comments()
{
    return $this->hasMany(Comment::class);
}

// Автор задачі
public function author()
{
    return $this->belongsTo(User::class, 'author_id');
}

// Призначений користувач
public function assignedTo()
{
    return $this->belongsTo(User::class, 'assigned_to');
}
}

```

User

```

<?php

namespace App\Models;

// use Illuminate\Contracts\Auth\MustVerifyEmail;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;

class User extends Authenticatable
{
    /** @use HasFactory<\Database\Factories\UserFactory> */
    use HasFactory, Notifiable;

```

```

/**
 * The attributes that are mass assignable.
 *
 * @var array<int, string>
 */
protected $fillable = [
    'name',
    'email',
    'password',
];

/**
 * The attributes that should be hidden for serialization.
 *
 * @var array<int, string>
 */
protected $hidden = [
    'password',
    'remember_token',
];

/**
 * Get the attributes that should be cast.
 *
 * @return array<string, string>
 */
protected function casts(): array
{
    return [
        'email_verified_at' => 'datetime',
        'password' => 'hashed',
    ];
}

```

```

    }

    public function projects() {
        return $this->hasMany(Project::class);
    }
}

```

Providers

AppServiceProvider

```

<?php

namespace App\Providers;

use Illuminate\Support\ServiceProvider;

class AppServiceProvider extends ServiceProvider
{
    /**
     * Register any application services.
     */
    public function register(): void
    {
        //
    }

    /**
     * Bootstrap any application services.
     */
    public function boot(): void
    {
        //
    }
}

```

Routes

Api

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProjectController;
use App\Http\Controllers\TaskController;
use App\Http\Controllers\CommentController;

Route::apiResource('projects', ProjectController::class);

Route::apiResource('tasks', TaskController::class);

Route::apiResource('comments', CommentController::class);
```

Console

```
<?php

use Illuminate\Foundation\Inspiring;
use Illuminate\Support\Facades\Artisan;

Artisan::command('inspire', function () {
    $this->comment(Inspiring::quote());
})->purpose('Display an inspiring quote')->hourly();
```

Web

```
<?php

use Illuminate\Support\Facades\Route;

Route::get('/', function () {
    return view('welcome');
});
```

```
Route::get('/myroute', function () {  
    return ('welcome');  
});
```